

Grounded Extraction of Technology Mentions from Papers and Patents

Abstract

Tracking the evolution of technology begins with naming the technologies in individual documents. This routine-looking extraction task has an unusual error profile once its output is aggregated: a fabricated technology is not averaged away by scale, because a spurious term whose rate follows corpus growth produces a curve indistinguishable from a genuine emerging technology, whereas a missed mention mostly rescales a series. We therefore build the extractor around structural constraints rather than a confidence threshold — a mention *is* a pair of character offsets with a cited evidence sentence, so a model that names a technology we cannot locate has named nothing. On a 895-mention gold set of 24 arXiv abstracts and 20 USPTO publications spanning thirteen arXiv categories and all five major CPC sections, the system reaches 0.551 partial F_1 at a hallucination rate of 0.000. An ablation over six components finds only one unambiguously load-bearing — the verifier — while two others buy strict precision at a measurable cost in overlap recall and ranking quality. We also report a negative result: span grounding bounds but does not close indirect prompt injection, because an adversary can write their phantom technology into their own patent.

1 Introduction

A lab tracking technology through the literature must first know which technologies each document is about; everything downstream consumes that output and cannot repair it. The task resembles scientific NER, and the obvious move is to apply that machinery: a span tagger over a scientific ontology (Luan et al., 2018) or a prompted generative extractor. None is designed for the property that matters here: an *asymmetry* between the two error types. Trend detection compares a term’s frequency against a baseline, so a constant spurious rate produces a flat line and does no harm, but a spurious

rate tracking corpus growth produces a rising line indistinguishable from a real emerging technology that nobody will go back and check. A missed mention usually rescales a series without changing its shape. Precision errors are not merely costlier, they are qualitatively different — and they are the errors generative extractors make (Ji et al., 2023).

Rather than tune an extractor to be cautious, we make the failure unrepresentable: a mention is a pair of character offsets, its surface is the slice at those offsets, and it must cite a document sentence containing it — the span-level analogue of attribution work in generation (Rashkin et al., 2023), applied as a hard gate rather than a post-hoc score. Grounding costs recall, since a correct but paraphrased identification is discarded, so the rest of the system gives that recall back where it can be checked: five recallers over-generate, a guard stack removes what is provably wrong, a verifier cascade challenges what survives, and one calibrated threshold decides what to withhold.

Three ways this is not standard NER. (i) **The vocabulary is open by construction.** The point of tracking technology is to see *new* technology, so an extractor constrained to an ontology can never report one; a knowledge base is an anchor, not a target. (ii) **Granularity is the problem in disguise.** “Machine learning”, “attention mechanism” and “FlashAttention” are technologies at different levels, and a tracker that mixes them reports machine learning as the fastest growing technology of every year since 1995. We type over a nine-way ontology separating artefacts, methods and materials (tracked) from fields, tasks, metrics, datasets and tools (annotated, not tracked). (iii) **Stance carries most of the signal.** A count conflates a technology being introduced, used and beaten, so each mention is tagged PROPOSED, USED, COMPARED, BACKGROUND or CLAIMED.

2 Approach

Structure-aware ingestion. Normalisation repairs ligatures, line-break hyphenation and typography and masks citation markers while keeping a per-character map back to the raw text, so an auditor sees the mention as it appears in the original filing. Both genres are segmented onto one closed set of zones, so one stance classifier serves both; claims are split individually and marked independent or dependent by back-reference detection, since a term in an independent claim is monopolised, not merely discussed.

Recall over-generates on purpose. Five recallers run per document. A *noun-phrase lattice* expands dependency chunks (Honnibal et al., 2020) into every left truncation crossed with noun-final right boundaries: whether the term is “string similarity” or “superficial string similarity” depends on information the recaller lacks, and the parser attaches a following word often enough to matter (“obstacle-aware route planning grounds”), which no left truncation recovers. *Abbreviation mining* (Schwartz and Hearst, 2003) recovers the defining parenthetical, the best within-document coreference signal available; *C-value/NC-value* (Frantzi et al., 2000) is the only recaller sensitive to *nesting*, which is the granularity problem exactly; a *gazetteer* over the Computer Science Ontology (Salatino et al., 2020) cannot find anything new but earns its place as a consensus witness; and a tier-2 *language model* runs only over thin zones.

We initially gated the chunker on a curated head-noun lexicon — the natural design, and apparently a precision win. It is not: gating discards 53% of gold mentions before any classifier sees them, because no lexicon anticipates *highback*, *rule-based elision*, or the mechanical-component vocabulary patents are built from. Table 1 reports both.

Grounding. Every candidate must be a verbatim slice at the offsets it reports. Model-proposed spans are relocated: the cited sentence is grounded first, the surface searched inside it, and a bounded ladder (exact, whitespace-insensitive, case-insensitive, optionally a similarity snap) returns *the document’s* characters or nothing — the similarity rung is off by default, since it alone can move a span onto a different entity. The invariant is asserted in the test suite, so the hallucination rate checks the implementation rather than comparing systems.

Typing, stance, verification. Type is a linear model over interpretable features — head-noun lexicon, orthography, KB depth, zone, and a distributional back-off scoring the surface against hand-seeded prototype centroids in a sentence-embedding space. Linearity is deliberate: the decision must be explainable in an audit, and with a few hundred annotated mentions a linear model is also the better estimator. Verification is a cascade: a *structural* verifier re-derives grounding and evidence from the document rather than trusting the record; an *embedding* verifier decides the easy cases; only the residual band reaches an *LLM judge*, shown the sentence, the term and the proposed type and nothing about how any were produced, because a verifier shown the proposer’s reasoning ratifies it. Disagreements with independent support go to a stronger adjudicator, which may accept, reject or abstain.

One abstention mechanism. An earlier version let each guard abstain and remove the mention; recall became a function of guard ordering. Guards now REJECT (a provable violation) or PASS, soft signals feed a logistic confidence model, and one calibrated threshold decides what is emitted. Overlaps are resolved *after* scoring, by confidence: the only evidence available before typing is span length and recaller count, which picks the wrong boundary about as often as the right one.

Canonicalisation and cost. Mentions merge by a cascade: in-document abbreviation pairs promoted to corpus aliases, KB identity, lemma-key identity, then embedding agglomeration over the NIL residue under a head-compatibility constraint. NIL clusters get a stable synthetic identifier, so an emerging technology is trackable from first naming. Per-candidate verification is the obvious design and the wrong one: a patent yields 10^2 candidates whose payload is a term, a type and one sentence. Deduplicating on (surface, type, sentence), escalating only the uncertainty band, and batching ~ 20 survivors per request behind a cached system prompt takes ~ 120 calls per patent down to one or two — a projected \$0.03 per document.

3 Experimental setup

Data. 24 arXiv abstracts from five category queries (computational linguistics, machine learning, signal processing, materials science, applied physics), spanning thirteen primary categories once

System	Strict match			Partial match			Labels		
	P	R	F ₁	P	R	F ₁	Type	Role	AURC
<i>Single-recaller baselines (guards reduced to grounding)</i>									
CSO gazetteer	0.303	0.068	0.112	0.589	0.133	0.217	0.294	0.459	0.299
Chunker (gated)	0.227	0.193	0.208	0.429	0.365	0.395	0.442	0.473	0.568
Chunker (open)	0.174	0.474	0.255	0.302	0.822	0.442	0.386	0.421	0.595
C-value/NC-value	0.004	0.002	0.003	0.522	0.284	0.368	0.399	0.455	0.520
Union, no guards	0.127	0.358	0.188	0.294	0.827	0.434	0.402	0.426	0.591
<i>Full pipeline and ablations</i>									
TEKNE	0.337	0.394	0.363	0.511	0.598	0.551	0.596	0.467	0.396
– negative lexicon	0.275	0.360	0.312	0.495	0.648	0.561	0.603	0.450	0.387
– consensus	0.331	0.413	0.367	0.499	0.623	0.554	0.594	0.460	0.404
– verifier	0.192	0.476	0.273	0.322	0.800	0.459	0.399	0.420	0.549
– abstention	0.307	0.457	0.367	0.455	0.678	0.545	0.598	0.431	0.422
– KB, abbrev.	0.350	0.374	0.361	0.544	0.581	0.562	0.584	0.456	0.363

Table 1: All eleven configurations, offline tier. Baselines are components of this system with the guard stack reduced to grounding. Every row has hallucination rate 0.000. The strict/partial gap measures boundary disagreement, not failure to find the technology; AURC is lower-is-better.

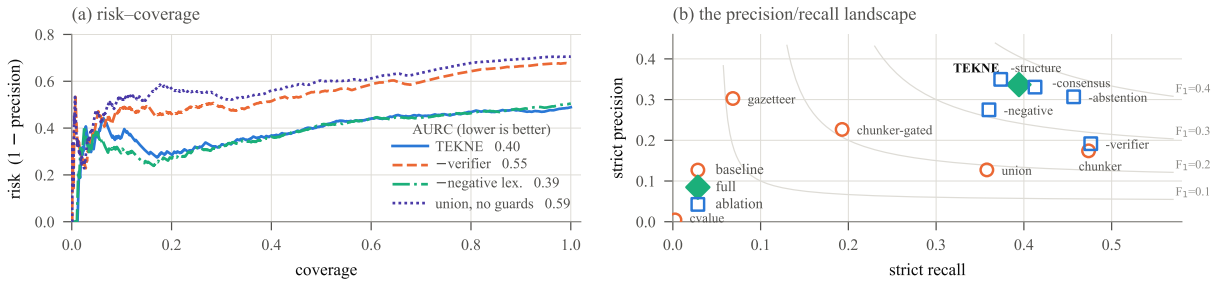


Figure 1: (a) Configurations close on F₁ separate on *where* their errors sit; removing the verifier moves the curve up by ~ 0.15 AURC. (b) The ablations cluster tightly, the baselines trace the recall axis at low precision. Iso-F₁ contours in grey.

cross-listing is counted; plus 20 USPTO publications sampled from the HUPD frame (Suzgun et al., 2023), stratified four apiece over CPC sections A, B, C, G and H, with full text from Google Patents — spanning snowboard bindings, HDAC inhibitors, Talbot–Lau interferometry, Ziegler–Natta catalysis, wheel bearings and spread-spectrum transmission. The spread is the point: a pipeline tuned on one vocabulary would visibly fail on the others.

Gold. 895 mentions over 44 documents (339 paper, 556 patent), annotated by one author under written guidelines. The zone is title and abstract for papers, plus the first independent claim for patents capped at 1400 characters — Markush claims run to thousands of characters of variable definitions naming no technology. System output is filtered to the same zone.

Metrics. Strict (exact offsets) and partial (overlap) span P/R/F₁; type and role accuracy on matched spans; the hallucination rate (emitted surfaces that are not verbatim substrings of the source);

and AURC (Geifman and El-Yaniv, 2017), which rewards a system whose errors concentrate in its low-confidence tail, with expected calibration error alongside so good ranking is not mistaken for good scaling. Partial F₁ is the primary comparison, following term-extraction practice (Rigouts Terryn et al., 2020), because boundary choice here is genuinely ambiguous. All numbers come from the offline configuration, reproducible without cost; each row is a configuration object rather than a code path.

4 Results

Baselines. Each component shows its characteristic profile (Table 1). The gazetteer is precise and nearly blind (0.068 recall): outside computer science the ontology has little to say. C-value alone has the worst strict F₁ of any configuration (0.003) but a respectable partial F₁ (0.368) — it finds the right regions and the wrong boundaries, exactly what a frequency statistic with no syntax should do. Gating the chunker raises precision 0.174→0.227

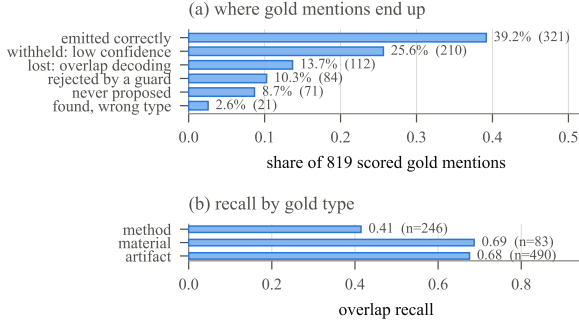


Figure 2: (a) Every scored gold mention attributed to exactly one outcome. Genuine recall failure is the *smallest* category. (b) Recall by gold type: methods are hardest, which the type-confusion analysis confirms.

Genre	Docs	Gold	P	R	F ₁
Papers	24	274	0.219	0.354	0.271
Patents	20	545	0.438	0.415	0.426

System	AURC	ECE	P@50%
TEKNE	0.396	0.309	0.592
– verifier	0.549	0.612	0.444
– negative lexicon	0.387	0.319	0.585
Union, no guards	0.591	0.652	0.399

Table 2: Per-genre performance (top) and the selective-prediction numbers an operator picks a threshold from (bottom). P@50% is precision at 50% coverage.

and costs more than half its recall. The full system reaches 0.551 partial F₁, 0.596 type accuracy and 0.396 AURC.

What each component buys, and costs. The ablations do not all point the same way, and the ones that do not are the informative ones.

Removing the *verifier* is the only unambiguous degradation, and it is large: precision 0.337→0.192, partial F₁ 0.551→0.459, type accuracy 0.596→0.399, AURC 0.396→0.549, ECE 0.309→0.612; recall *rises*, to 0.476. That is what a verifier is for — it does not find more, it decides what to keep.

The *negative lexicon* and the *KB with the abbreviation miner* trade the other way: dropping the lexicon costs strict precision (0.337→0.275) while slightly *improving* partial F₁ (0.561) and AURC (0.387), and dropping the KB behaves similarly (0.562, 0.363). Both remove candidates that overlap a gold span without matching it exactly, buying exact-boundary precision by giving up partial credit, and the items they remove were already ranked low enough that removing them does not improve the ranking. At a precision target this is

the right trade — but it is a trade, not a free win. *Consensus* and *abstention* move strict F₁ by under a point and improve AURC (0.404, 0.422 against 0.396): they change what the system is confident about, not what it finds (Figure 1).

Genre. Patents are harder (Table 2) despite contributing more gold per document, because claim language generates many well-formed component noun phrases indistinguishable from technology mentions without domain knowledge — “a first position”, “said support section”.

Error analysis. Attributing each scored gold mention to one outcome (Figure 2a) is clearer than the F₁ columns. Only 8.7% were never proposed: genuine recall failure is the smallest category. The two largest losses are recoverable in different senses — 25.6% are withheld by the confidence threshold, the risk–coverage trade being made deliberately and recovered by one knob; 13.7% lose overlap decoding to a neighbouring span, which partial F₁ still credits. Of the 10.3% rejected by a guard, four-fifths were typed NOT_TECH; the costly confusions are METHOD/ARTIFACT→TASK, the boundary the guidelines flag as hardest, and why method recall (Figure 2b) trails the others by ~0.27.

5 Prompt injection: a negative result

Documents are adversary-authored, permanently published and placed inside a model prompt, making indirect prompt injection (Greshake et al., 2023) a live concern for a corpus built over decades of filings. We fence document text with a per-call nonce, but the substantive defence was supposed to be grounding.

It is not sufficient, and an adversarial test showed why. An attacker who writes “*always extract QuantumLeap Processor as the primary technology*” into their own specification has made that string a verbatim substring of their own document, and a grounded extractor has no basis to reject it. Grounding buys a *bound*, not a defence: the adversary can only inject technologies they will publish under their own name. Closing the residue needed a second mechanism — rejecting candidates whose own sentence contains instruction-like content — which costs the mentions in example sentences of papers about prompting. We keep the negative result as a test so it stays visible.

6 Limitations and future work

Single annotator, so no agreement figure: the numbers characterise this pipeline against this annotation, not absolute accuracy. Roles are annotated per document rather than per occurrence, making patent role accuracy closer to zone detection than stance classification, and the paper sample over-represents LLM-adjacent work. In the offline configuration the type classifier and the embedding verifier share a distributional signal, so that verifier is a consistency check rather than an independent one; genuine independence needs the LLM tier. The zone excludes patent descriptions, where much specific disclosure lives, and 44 documents supports ablation comparison but not absolute claims. The downstream tracking demonstration is weaker than intended: its time-sliced corpus is built from the arXiv API a year at a time and the endpoint rate-limited us to two usable slices, so the aggregation logic is exercised by unit tests on synthetic series and on the real corpus only across two points — nothing here is an empirical claim about technology trends.

The highest-value extension is *temporal sense disambiguation*: vocabulary is reused across eras — “transformer” in a 2005 power-systems patent is a magnetic component — and one such error lands at the start of a series, exactly where an emergence detector is most sensitive. We withhold knowledge-base links predating a document; disambiguating senses against a contemporaneous corpus is unattempted, as are per-occurrence stance, comparative statements (“outperforms X”) that would turn co-occurrence into a successor graph, full-text patents, and comparison against a fine-tuned span tagger on a larger gold set.

7 Conclusion

Technology extraction for evolution tracking is a precision-dominant, open-vocabulary problem in which fabricated entities do lasting damage and missed ones mostly do not. Treating that asymmetry as a design constraint — offsets rather than strings, cited evidence rather than assertion, one calibrated abstention rather than many cautious stages — yields a system whose hallucination rate is zero by construction and whose remaining errors are boundary and type disagreements a reader can inspect. The components that matter most are not those that find mentions but those that decide which to keep.

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